

Simplified Notes: TLS, SSH, and VPN

1. TLS with Common Protocols

TLS secures many protocols just like HTTPS. Adding TLS makes them secure:

Protocol	Insecure Port	Secure Port (TLS)
HTTP	80	443 (HTTPS)
SMTP	25	465 / 587 (SMTPS)
POP3	110	995 (POP3S)
IMAP	143	993 (IMAPS)

2. Secure Shell (SSH)

- Invented in 1995 to replace insecure Telnet (which sent cleartext passwords).
- OpenSSH (1999) is the most used implementation today.
- **Benefits:**
 - Secure authentication (password, public key, 2FA)
 - Encryption for confidentiality
 - Integrity checks
 - Tunneling (acts like VPN)
 - X11 forwarding for GUI apps
- Command example: `ssh username@hostname`

3. Virtual Private Network (VPN)

- Connects remote offices/branches securely over the Internet.
- Creates an encrypted tunnel (protects confidentiality & integrity).
- Economical solution using existing Internet infrastructure.
- Needs: VPN client + VPN server.
- Example: Two branches connect to main office VPN server. Encrypted traffic (blue lines) moves securely, decrypted inside the network (green lines).

Extra Tips for Practice

- In Wireshark: Always set the SSL key log file under TLS preferences to decrypt HTTPS/TLS.
- Look for **POST requests** with `application/x-www-form-urlencoded` content-type for login data.
- Expand HTTP/2 headers carefully — credentials often appear in the payload/body.
- Compare encrypted vs decrypted traffic to understand what TLS protects.
- Remember: **Telnet = insecure**, **SSH = secure**, and **VPN = secure tunnel**.